ASIA CCS '26

Uncovering the Deceptive Tactics of Stalkerware

A Large-Scale Measurement Study

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Dual use apps don't get the same scrutiny as stalkerware



The Conversation

Location-sharing apps are enabling domestic violence. But young people aren't aware of the danger

n young people's perceptions of technology-facilitated abuse has shown re not aware of the danger.



He also tracked [redacted]'s location on Snapchat to monitor her whereabouts and asked a mutual friend to keep "an eye on her" during a party she attended.

What they do alone **doesn't separate them.**
So we ask *what they say*.

Research Design



Stalkerware: apps that covertly monitor a victim's phone, usually without consent.

ALREADY STUDIED

What these apps can do

Their harms, and how to detect them

NOT MEASURED

RQ1 Disclosure Misalignment (Stalkerware vs. Benign Apps)

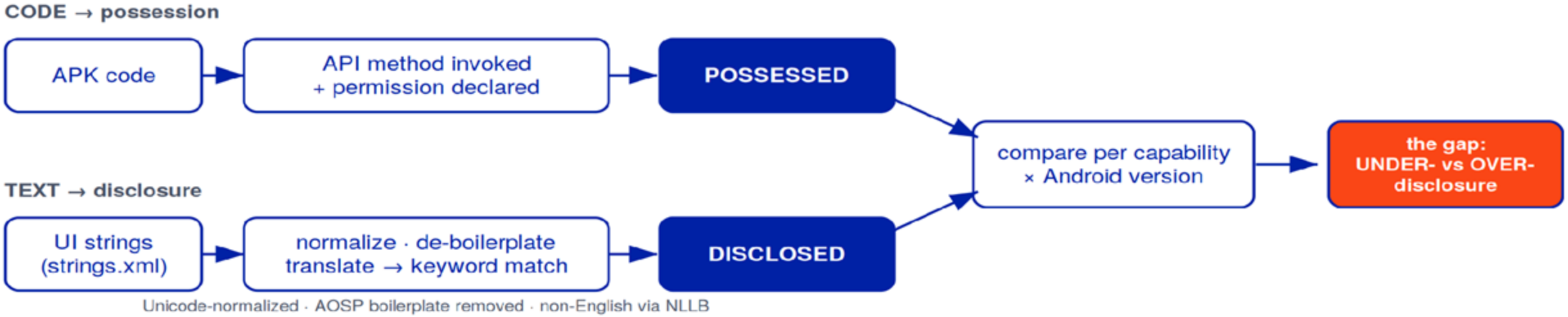
RQ2 Variation within Stalkerware and Dual Use apps

RQ3 Evolution of what the apps say

We measure that gap: 8,427 samples across 9 Android versions.

Methodology

How we measure the gap



TEXT PATH → DISCLOSED

"Track your kids location and keep them safe..."
discloses one capability: location

CODE PATH → POSSESSED

Location LocationManager API + ACCESS_FINE_LOCATION
SMS SMSProvider API + READ_SMS
Call recording MediaRecorder API + RECORD_AUDIO
Screen capture Accessibility Service API + BIND_ACCESSIBILITY_SERVICE ·

Discloses 1 of 4: three under-disclosures, zero over.

Corpus & validation

Stalkerware corpus from the Coalition Against Stalkerware threat list & benign baseline from F-Droid.

8,427 stalkerware samples

9 Android versions

11 surveillance capabilities

645 benign baseline

Possessed capability detection (n = 75)

Capability	Precision	Recall
Location	98.6%	100%
Calendar	100%	87.5%
SMS	98.1%	89.5%
Call logging	96.8%	98.4%
Call recording	100%	80.9%

Disclosed capability detection (n = 100)

Capability	Precision	Recall	F1
Location	100%	94.7%	97.3%
Calendar	100%	95.8%	97.8%
SMS	100%	93.1%	96.4%
Call logging	100%	94.6%	97.3%
Call recording	97.3%	92.3%	94.7%

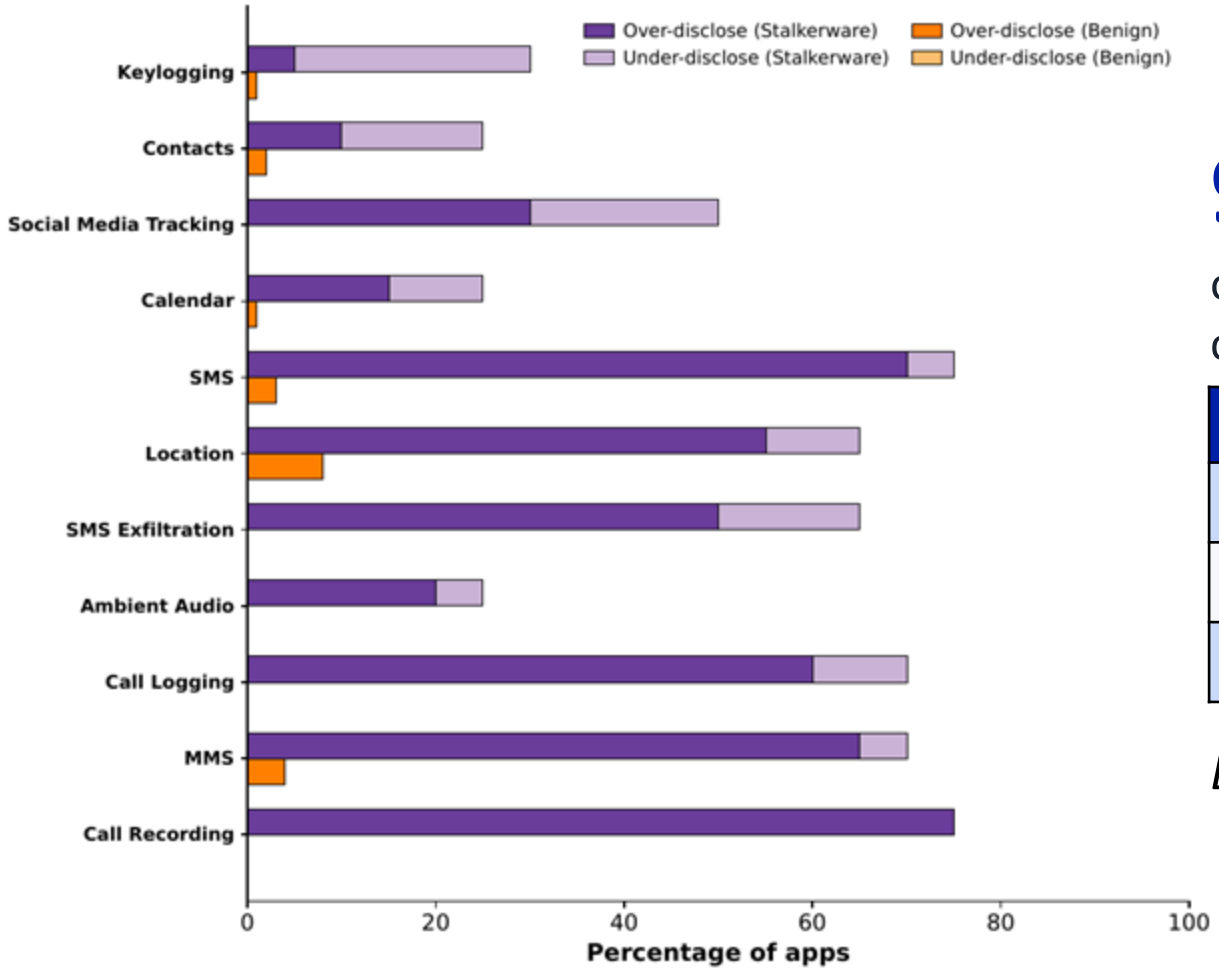
Misalignment is the norm, not the exception

98.6% of stalkerware apps show disclosure misalignment.

Almost entirely concealment: 97% under-disclose a capability they actually have.

Measured across 7,377 stalkerware apps and 9 Android versions.

RQ1



The gap is real, and huge

97% vs **1.55%**

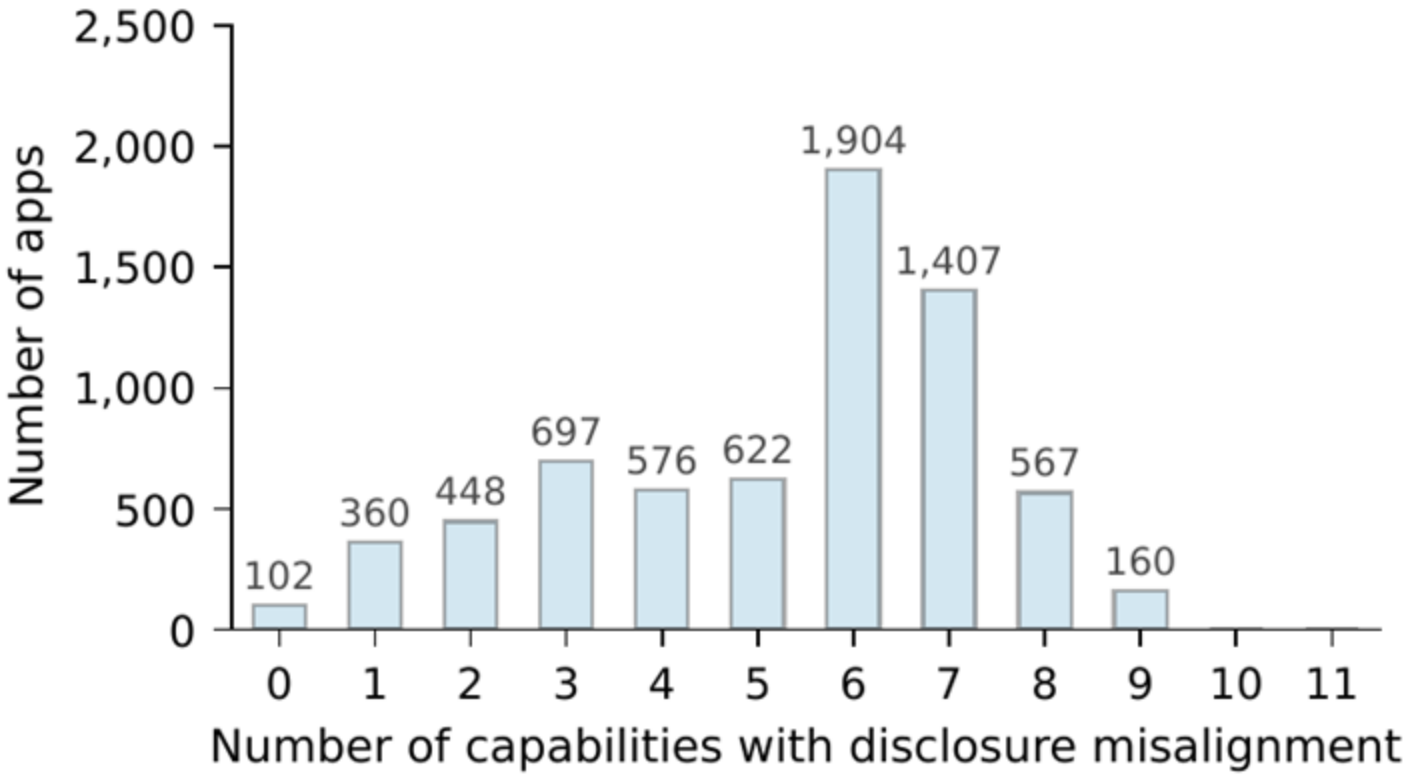
of stalkerware vs benign apps under-disclose at least one capability

	Stalkerware	Benign
Call recording	**76.3%**	0.2%
SMS	**71.1%**	7.3%
Location	**59.1%**	**10.1%**

Benign stays near-zero on every other capability

RQ2

Misalignment varies throughout the corpus



How they brand themselves (qualitative)

0	"Parental monitoring"
4	"Employee monitoring"
6	"Family tracker"
7	"Spy app"
9	"Covert spy app"

RQ3

Some apps don't spy, they teach others how

9%

765 apps

request zero permissions.

They are installation guides for the actual apps.

Trigger none of the four signals that flag stalkerware:

- ✗ dangerous permissions
- ✗ data sent off-device
- ✗ known malware signature
- ✗ advertises as surveillance

TAKEAWAYS

01 The gap is real

98.6% vs. 14%. Not a marginal signal.

02 The gap varies

Clusters at 4–7 capabilities; appears to relate to positioning.

03 The gap is a candidate signal

Questions?

If you or someone you know is affected by stalkerware:

Coalition Against Stalkerware · stopstalkerware.org · how to detect, remove, find local help
TechSakhi (India) · call / WhatsApp 080 4568 5001 · techsakhi.in
India National Cyber Crime portal · cybercrime.gov.in · helpline 1930

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Slides for Q&A

Validation: our pipeline vs. FlowDroid

50 randomly-sampled apps, SDK 28/29/31

Capability	FlowDroid	Ours	Δ
Location	19	23	+4
Calendar	3	3	0
Contacts	17	18	+1
Call logging	7	7	0

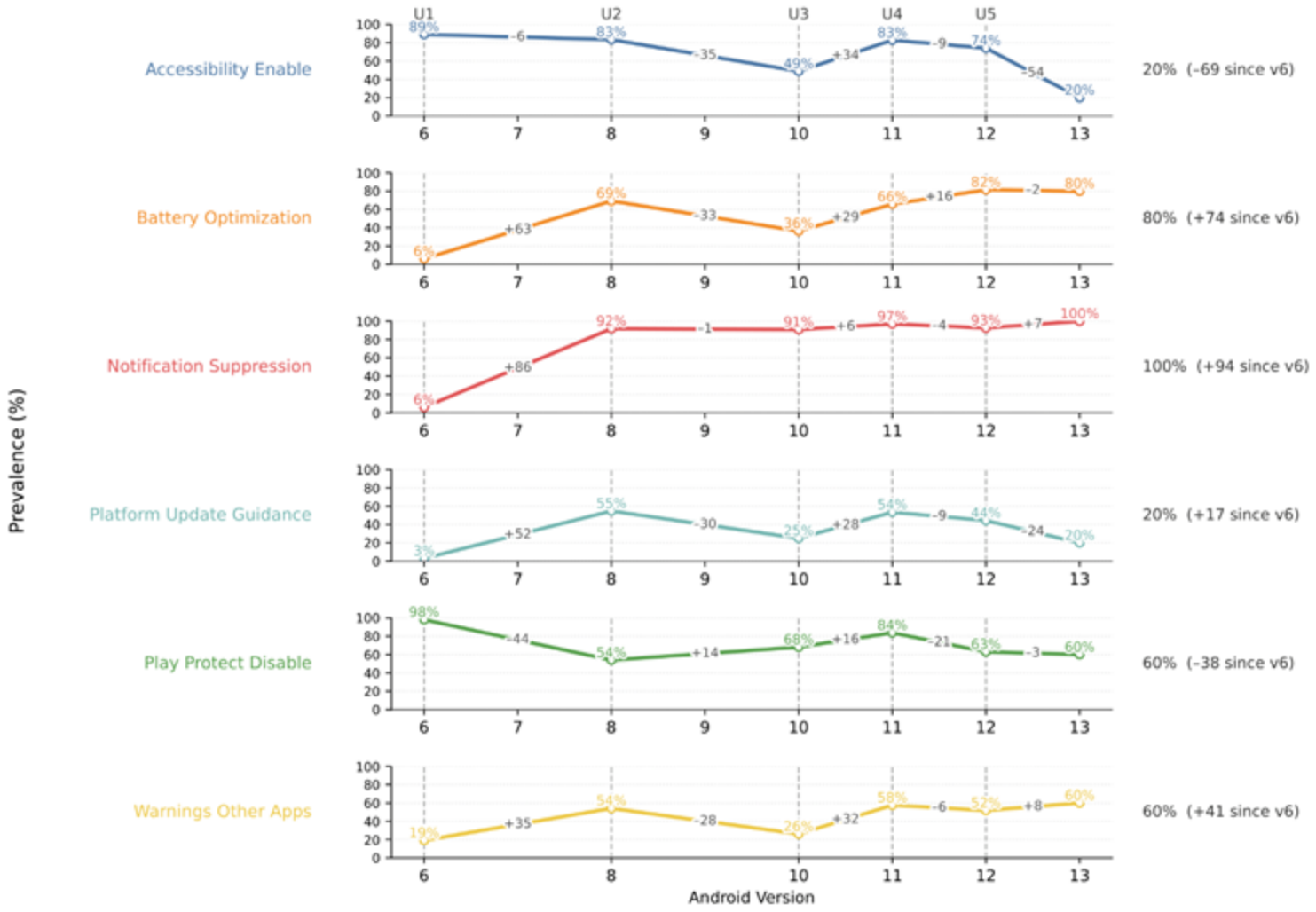
Why the gap?

FlowDroid missed 4 location + 1 contact use.

These apps use reflection-based calls, our regex + permission check catches them.

Keyword detector (n=100): precision 97–100% · recall 92–96% · F1 94–97%

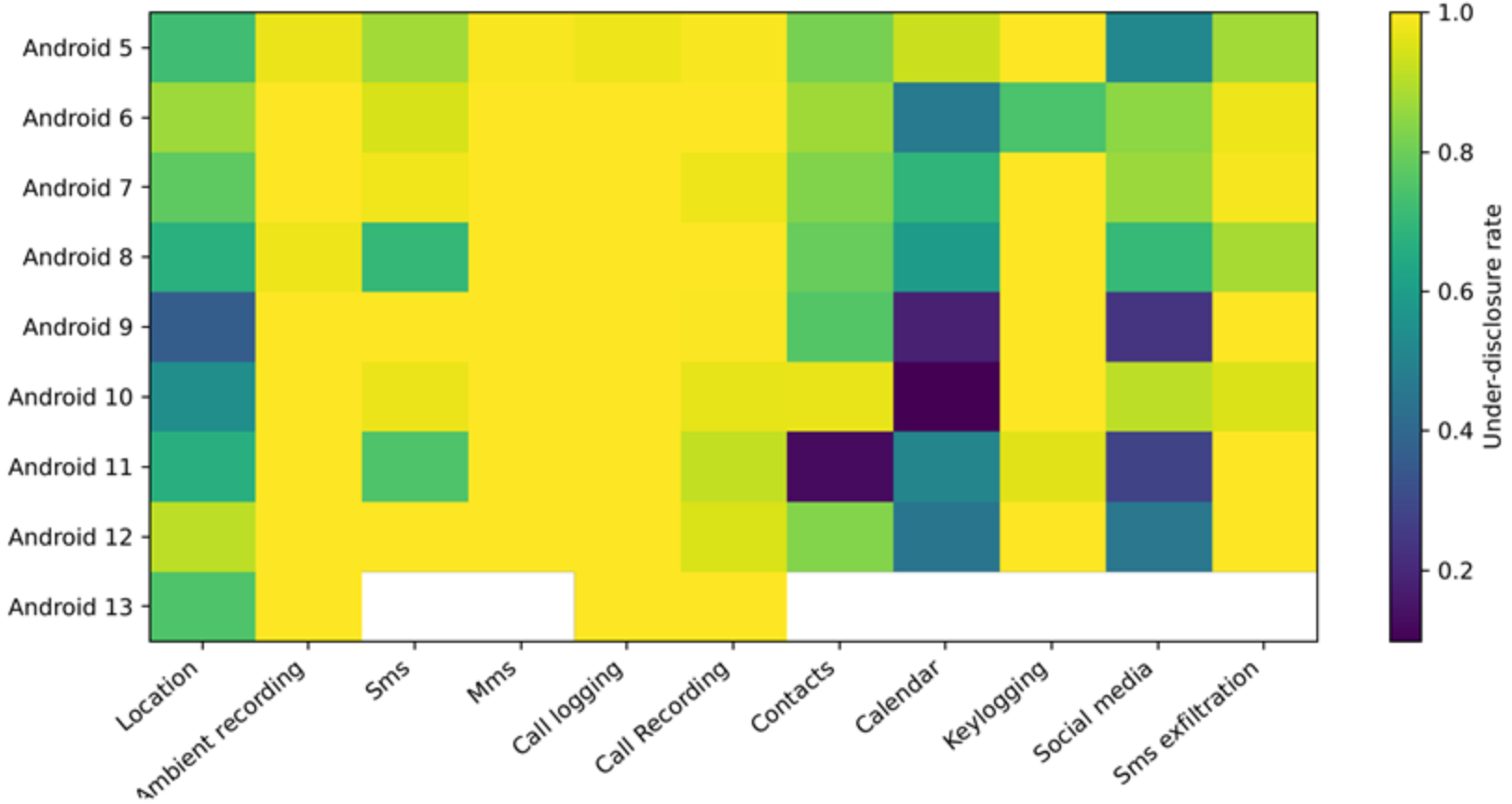
RQ3



Stalkerware evolves with Android

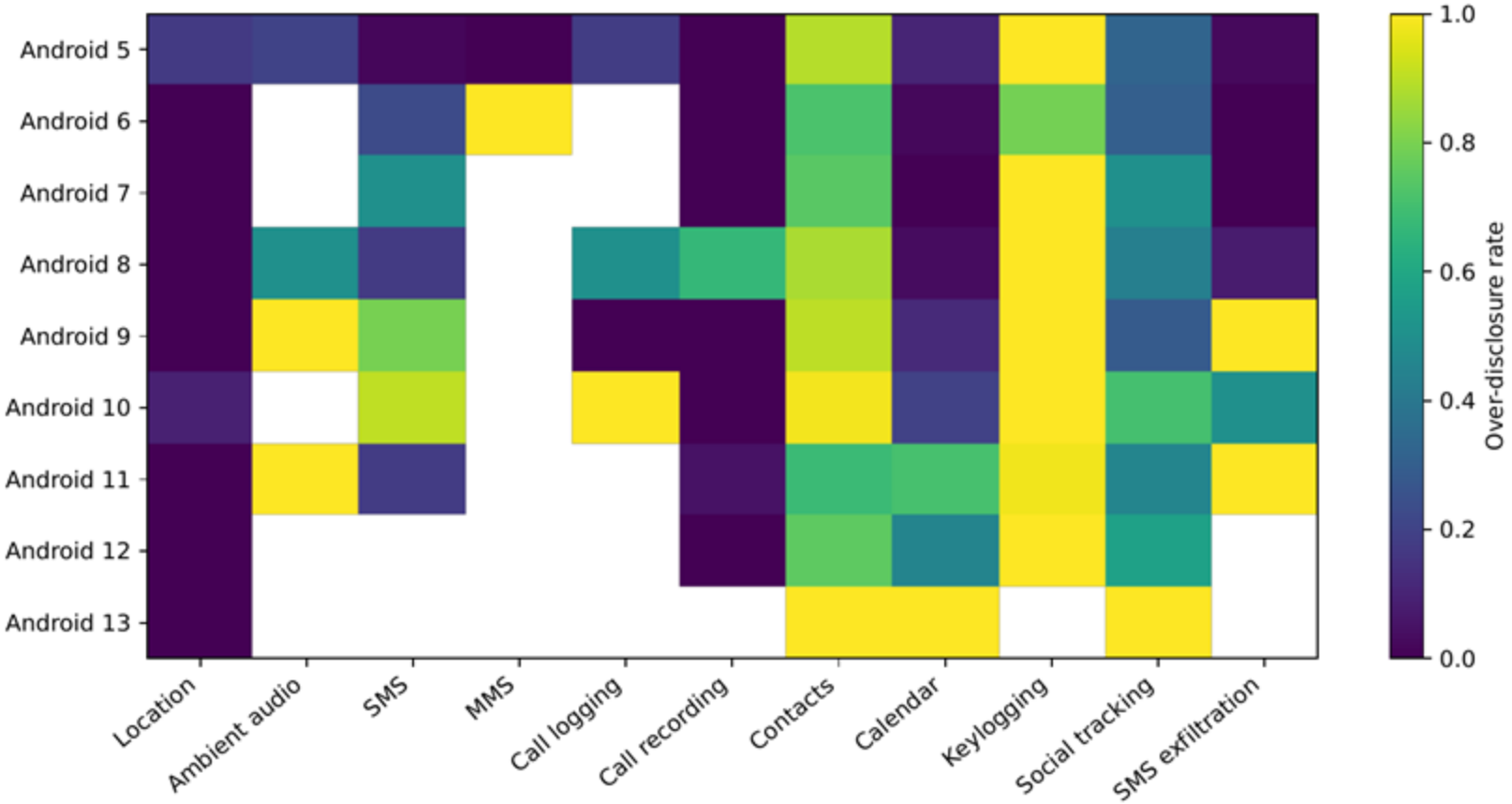
Installer strings detected with all-MiniLM-L6-v2 sentence embeddings; cosine ≥ 0.65 to seeded category actions.

Under-disclosure across Android versions



Values = % of apps that possess the capability but do not disclose it.

Over-disclosure across Android versions



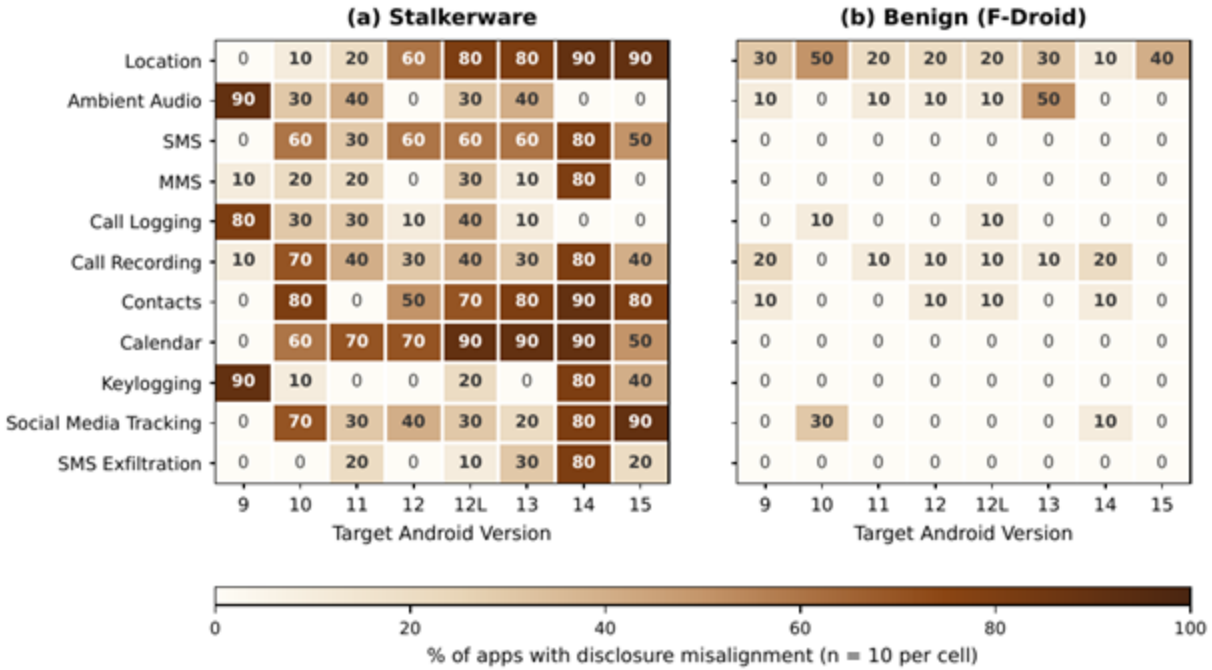
Values = % of apps that disclose the capability but do not actually possess it.

Installer-instruction categories

Category	% of apps	Example
accessibility_enable	65.9%	*"Enable accessibility service."*
battery_optimization	44.6%	*"Add the App to Protected apps list."*
notification_suppression	60.9%	*"You can hide this icon as follows..."*
platform_update_guidance	37.3%	*"Re-enable screen streaming after restart."*
play_protect_disable	68.9%	*"Turn off Play Protect to install."*
warnings_other_apps	41.1%	*"Uninstall any antivirus first."*

Detection: MiniLM sentence embeddings, cosine ≥ 0.65 vs. seeded examples.

Findings hold on Android 14–15



INSTRUCTIONS, 232 NEW SAMPLES

Notification suppression	**89.7%**
Battery optimization	**75.0%**
Accessibility enable	**72.0%**
Icon hiding	**63.8%**
Warnings other apps	**58.2%**
Play Protect disable	**50.9%**

Limitations and ethics

Static analysis only

Can't prove runtime execution. Results are a lower bound.

Keyword noise

Euphemisms may be missed; boilerplate may inflate. Manual review + emulator checks.

Intent not provable

Misalignment ≠ intent. We report the signal, not motive.

Mitigations

We don't name specific packages. Available to researchers on request.